

Optimal use of electric energy oriented water-electricity combined supply system for the building-integrated-photovoltaics community

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HIGHLIGHTS

- A water and electricity combined supply system in a smart community is proposed.
- Shapley value method is firstly introduced to distribute the cooperation revenues.
- Forecast errors are addressed by a two-stage electric and hydraulic combined method.
- The iterative procedure can solve the hydraulic and electric dispatching problem efficiently.

ARTICLE INFO

Keywords:

Building-integrated photovoltaics
Water distribution system
Shapley value method
Electric-hydraulic combined dispatching
Iterative method

ABSTRACT

As a new application form of sustainable energy generation, building-integrated photovoltaics provide great flexibility for energy conservation and emission reduction. However, due to the randomness and volatility from photovoltaic generation, the full exploitation of the advantages of building-integrated photovoltaics has become an important question for the utilities. To address this issue, this paper proposes a novel and comprehensive (combined) water-energy supply system applied to the smart community with building-integrated photovoltaics and two-stage electric-hydraulic dispatching method. It was designed to minimize the electricity consumption cost and power fluctuation while meeting the water demand loads, followed by the iterative algorithm as well as linearized techniques to solve the optimisation model. The Shapley value method was introduced to ensure the fair revenue between the property company and owners. Simulation studies on a building-integrated-photovoltaics community and a 15-node water distribution system indicates that the annual operation cost of the water-electricity combined supply system was reduced by 29.6% while the external grid obtained a better prediction regarding the real-time exchange power. The results confirm the effectiveness of the proposed model on peak shaving and valley filling and fluctuation smoothing.

1. Introduction

The current energy crisis and environmental requirements have stimulated the development of building-integrated photovoltaics (BIPV) [1]. However, BIPV applications must deal with the problem of the owners bearing an economic loss due to either PV power curtailment or selling to the external grid (EG) at a low price during the time periods when the photovoltaic (PV) peak output exceeds the load demand [2]. Meanwhile, the unpredictability and volatility characteristics of PV give rise to great challenges in achieving power balance and reliable microgrid (MG) operation [3]. Although the above problems can be solved well by electric energy storage systems, their application has been limited by their high operational and maintenance costs and other

drawbacks [4]. Thermal energy storage systems such as residential buildings with air conditioning [5], wall bricks with a heater [6] or hot water storage with a heat pump [7] are regarded as economic solutions, but the weak heating/cooling demand in regions with perennial comfortable temperature leads to a limited annual comprehensive utilization rate. Therefore, to fully exploit to the advantages of BIPV, the development of an economical and universal energy storage system is urgently needed.

On the other hand, as it is well known that the water treatment acts as a common and considerable consumer, thus many studies have focused on its electricity consumption and sought breakthroughs in the energy conservation. For instance, it occupied approximately 3% of the total electricity in the United States [8] and 20% of that in California

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<https://doi.org/10.1016/j.apenergy.2019.04.059>

Received 4 December 2018; Received in revised form 20 March 2019; Accepted 14 April 2019

Available online 19 April 2019

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[9] and 30–40% of that consumed by municipal governments [10]. According to the estimation of the World Bank, the above proportion will increase significantly due to the growing water demand at an expected rate ranging from 43% in North America to 283% in Sub-Saharan Africa from 2005 to 2030 [11]. However, unlike other rigid large energy consumers, water distribution systems (WDS) incorporated water pumps (WPs) with the flexible adjustment capability for improving the efficiency and providing various services. For example, the WP was optimized by shifting electricity consumption to low-price periods to decrease its operational costs in [12]. Considering the rainfall and water demand uncertainties, [13] proposed a robust two-stage strategy to optimize the WP power. On this basis, [14] introduced an online robust control method for practical application and [15] carried out the multiple experiments and analysed a corresponding systematic error for improving reliability of the test results. In addition to the above works, several mathematical and heuristic optimization algorithms were introduced accordingly. Ref. [16] presented an algorithm for WP modelling and scheduling. It uses the WP start/end run times as continuous variables and the WP status at the beginning of the scheduling period as binary variables. On this basis, [17] introduced a two-stage solution method for the optimal scheduling problem of WDS where the relaxed continuous variables are optimized in the first stage and the binary variables are processed in the second stage by using a local branch and bound method. In [18], the operational optimization of pumping station with multiple WPs was treated as a dynamic programming problem and was solved using the reduced dynamic programming algorithm. Ref. [19] adopted a neutral evolutionary search method to solve the least cost WP operation optimization problem for an industrial size WDS. Six multi-objective evolutionary algorithms including strength Pareto evolutionary algorithm, multiple objective genetic algorithm and so on were introduced to solve the optimal WP-scheduling problem and compare the solution performance [20]. The operation optimization problems and solutions of WDS were comprehensively reviewed in [21].

The above-mentioned literature mainly focuses on the primary WDS operated by the water supply corporation in which the flexible operation of WPs can benefit the operators. However, for the secondary WDS (such as in a community) owned by a property company, in addition to WPs, water tanks are generally used to ensure safe and adequate water supply to the owners and these WDS have rarely attracted the attention of academic researchers. In practice, by optimizing WPs and tanks, the secondary WDS purchases electricity at a low price from the EG to reduce its operational cost as much as possible regardless of the MG, which means that the problems caused by the BIPV still exist. However, the perfect regulation effects of secondary WDS indicate that the community with BIPV can consume the excessive energy generated during PV peak output by storing the water in a tank and mitigating the power fluctuation by regulating the WPs; this approach has not yet been applied or mentioned in the literature. To elucidate the problems and verify the performance characteristics of these approaches, a novel water and electricity combined supply system (WECSS) involving different stakeholders, together with a detailed hydrodynamic model incorporating water tanks is proposed for the first time in this paper.

The existing work focusing on the linkages between the water and electricity industries to find opportunities to reduce energy consumption and operating costs provide an excellent reference for the modeling of WECSS [22]. For example, a multi-plant real-time simultaneous economic dispatch of power and water was developed to address the coupling relationship between WDS and the power grid in [23]. A quantitative engineering system of the energy–water nexus was modelled using bond graph representations of the system's modelling language in [24]. Ref. [25] developed a day-ahead power systems operation model that integrated the energy regulation flexibility of the WDS. Ref. [26] introduced an energy efficiency evaluation strategy for WDS design and a mathematical model for optimizing WP operation in high-rise buildings. Ref. [27] presented the energy-water optimal scheduling

model using harvested rooftop rain water and the time-of-use electricity price tariff. A framework consisting of two optimization models was proposed in [28] where the first-step model optimizes the procurement cost of the WDS operators and the second-step model optimizes the demand response and frequency regulation. Furthermore, with different peak price of the electricity tariff, the model that optimizes variable speed WPs in WDS to provide the demand response for a power system was established in [29]. Models that focus on passive energy efficiency and price-dependent load management methods in WPs and water treatment plants were presented in [30]. In [31], the energy use and the state of demand response readiness in wastewater treatment facilities were characterized. Although the long-time scale performance of WDS has been well examined in the above literatures, the flexibility provided by water tanks and WPs on a short-time scale such as by eliminating power fluctuations has not been modelled and studied. Additionally, concentrating on the WDS and large power grid, these models neglect the electric limits of the power flow and WP ramp rate and the hydrodynamic constraints of WDS for which the dispatching strategies may not be supported in WECSS involving WDS, MG and EG particularly under risk conditions. Moreover, as the basic premise for promoting cooperation, the distribution of the cooperation revenues among the participants with different interests and price barriers has not been discussed to date.

In addition to supplementing the shortcomings of existing work in modelling, this paper also proposes a two-stage hydraulic and electric combined dispatching method to ensure the performance of the WECSS on multiple time-scales. In fact, the dispatching problem has been a hot research topic [32]. For example, economic power generation strategies are formulated in the day-ahead stage while prediction errors are mitigated in the real-time stage in [33]. On this basis, [34] integrated model predictive control (MPC) into the two-stage dispatching model, significantly enhancing the ability of the dispatching system to cope with risks. Regardless of the violent fluctuation of real-time power, [35] directly adopted a one-stage MPC strategy to minimize the operational cost. Likewise, [36] used a hierarchical iterative control algorithm for optimal balance of the grid load (both the voltage drop and current load) while satisfying users power demand. By contrast, a power smoothing objective adopted in [37] resulted in the poor operation economy of an energy storage system. However, unlike the existing dispatching system, for WECSS, both electric balance and hydraulic balance in multiple stages should be met to simultaneously ensure safe water and electricity supply to residents. More importantly, the prediction errors such as the PV output error will have a profound impact on the electric and hydraulic balance on multiple time-scales, further increasing the dispatching difficulty of WECSS. Consequently, a novel dispatching method must be designed to coordinate the hydraulic and electric systems and eliminate the impact on multiple time-scales caused by prediction errors. Meanwhile, the method for solving the novel dispatching model of WECSS efficiently to meet the actual demand of the novel dispatching system must be carefully considered.

Summarising, to solve the storage problem caused by BIPV, a novel combined water-energy supply system is proposed. A detailed comprehensive model with technical considerations, constraints and its profit distribution is presented for the first time, to analyse its performance on multiple time-scales. (This is the special consideration and the main contribution in this paper.) In addition, the Shapley value method is introduced firstly to distribute the cooperation revenues between both sub-systems. Lastly, to reduce the impact of forecast errors, a two-stage dispatching method is proposed and an iterative procedure as well as linearized techniques is adopted to solve the problem efficiently.

The rest of this paper is organized as follows: Section 2 presents the materials and methods involving the structure of the water-energy supply system, the electric-hydraulic combined model, the two-stage dispatching method, case study and parameters and the multiple integer linear programming formulation. Results and discussion are

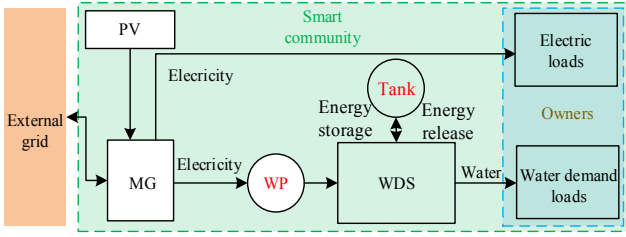


Fig. 1. Structure of the WECSS applied to a smart community with BIPV.

conducted in Section 3. Finally, the conclusions are drawn in Section 4.

2. Materials and methods

2.1. Structure of the water-energy supply system

With BIPV and WDS in a smart community, the designed WECSS is shown in Fig. 1. As observed from the figure, the electric-hydraulic systems are coupled with WPs. The MG connected to the EG can not only supply power to various electric loads, but also supports the power demand of WPs directly. The water demand loads are met by WDS service. There exist two operation modes of WECSS in the process of exchanging power with EG:

- Mode 1: Water storage mode. Electric energy is converted into potential energy of water by WPs and is stored in the tanks;
- Mode 2: Water release mode. Water with high potential energy in the tanks is released to reduce the power demand of WPs.

Of course, a reasonable revenue distribution method is the first prerequisite of the cooperation. To promote and maintain the cooperation between the property company and the owners, the Shapley value method used in economics is introduced in this paper to distribute the revenues equally according to their contributions [38]. The theory is as follows:

Assume that set $N = \{1, 2, \dots, n\}$ indicates that n members are engaged in economic activities. $V(S)$ expresses the revenues of cooperative S (a subset of N) and is represented by:

$$\begin{cases} V(\phi) = 0 \\ V(S_1 \cup S_2) \geq V(S_1) + V(S_2) \quad S_1 \cap S_2 = \phi \end{cases} \quad (1)$$

where V is the strategy eigenfunction. $\Phi = (\Phi_1(v), \Phi_2(v), \dots, \Phi_n(v))$ expresses the revenue distribution vector where $\Phi_i(v)$ denotes the revenues belonging to member i :

$$\Phi_i(v) = \sum_{i \in S} w(|S|)(v(S) - v(S \setminus i)), \quad i = 1, 2, \dots, n \quad (2)$$

$$w(|S|) = (n - |S|)! (|S| - 1)! / n! \quad (3)$$

where $|S|$ and n indicates the number of the subsets and members, respectively, and $v(S \setminus i)$ denotes the contribution of member i in S .

In this paper, water independent supply mode (WISM) for the property company and electricity-independent supply mode (EISM) for the owners and WECSS are denoted as

$S = (\{WISM\}, \{WISM\}, \{WECSS\})$ in which the corresponding revenue space can be expressed as $V = (V(\{WISM\}), V(\{EISM\}), V(\{WECSS\}))$, and the distribution vector is $\Phi = (\Phi_1(v), \Phi_2(v))$.

2.2. Electric and hydraulic combined model

The solar radiation, power constraints and dynamic structure of WDS (for instance, the pressure head) have a significant impact on the performance of WECSS in both the long and short term. As a result, a precise electric and hydraulic combined model incorporating the power flow limits and WP ramp rate was established to evaluate the working

characteristics of WECSS on multiple time-scales.

First of all, power sources modeling of PV and the EG were presented. Define P_{PV} as the output power of PV and G as the solar radiation at the surface of a photovoltaic panel (W/m^2). The output power of PV is controlled by the solar spectrum and can be represented as follows:

$$P_{PV} = G/1000 \times P_{PV, rated} \times \eta_{MMPT} \quad (4)$$

where $P_{PV, rated}$ is the rated power of the PV array at $G = 1000 W/m^2$ and η_{MMPT} is the efficiency of the PV's DC/DC converter in the Maximum Power Point Tracking System.

A smart community can purchase electricity from an EG at a price $p_{in,t}$ and sell electricity to the EG at a price $p_{out,t}$. Assume that the power exchanged between the EG and the community is $P_{EX,t}$ including the sold power $P_{EX,t}^{out}$ and the purchased power $P_{EX,t}^{in}$. The exchanged power satisfies the following constraint:

$$P_{EX} \leq P_{EX,t} \leq \overline{P_{EX}} \quad \forall t \quad (5)$$

where $\underline{P_{EX}}$ and $\overline{P_{EX}}$ are the upper and lower limits of the power exchanged between the community and the EG.

Secondly, the WECSS model on the electric side with the ramp rate constraints was established. The vectors of the WPs' power at time t are defined as $\mathbf{P}_t^W = (P_{t,1}^W, \dots, P_{t,U}^W)^T$. The power constraints are given by:

$$0 \leq \mathbf{P}_t^W \leq \overline{\mathbf{P}}^W \quad \forall t \quad (6)$$

$$\Delta \mathbf{P}^W \leq \mathbf{P}_t^W - \mathbf{P}_{t-1}^W \leq \Delta \overline{\mathbf{P}}^W \quad \forall t \quad (7)$$

where $\overline{\mathbf{P}}^W$, $\Delta \mathbf{P}^W$ and $\Delta \overline{\mathbf{P}}^W$ are the $U \times 1$ vectors of the maximum WP power, maximum down ramp rate and maximum upper ramp rate.

Finally, the WECSS model on the hydraulic side including the SOC (state of charge) definition was presented. The variables related to the hydrodynamic model shown in Fig. 2 are defined as follows: The external variable $\mathbf{Q}_t^D = [Q_{t,1}^D, \dots, Q_{t,N}^D]^T$ is the $N \times 1$ vector of the water load. The adjustable variables $\mathbf{I}_t$ and $\mathbf{O}_t$ are the $K \times 1$ vectors of the tanks' water input and output flows, respectively. State variables $\mathbf{V}_t$ and $\mathbf{SOC}_t$ are the $K \times 1$ vectors of the storage water and the corresponding energy storage state. To facilitate the understanding of the hydrodynamic Eqs. (8)–(21), the typical hydrodynamic network model consisting of a water pipe, WP, tank, source and loads is shown in Fig. 3 where the energy storage and energy release process are marked by the red and blue lines while black lines indicate the other flow exchange.

The dynamic balance between the water supply and demand is maintained at all times in WDS and can be expressed by the following formula:

$$\mathbf{Q}_t^R - \mathbf{Q}_t^D = \mathbf{A} \mathbf{Q}_t + \mathbf{\Lambda}^T [\mathbf{I}_t - \mathbf{O}_t] \quad \forall t \quad (8)$$

where $\mathbf{Q}_t^R$ is the $N \times 1$ vector of the source's water inflow, $\mathbf{Q}_t$ is the $J \times 1$ vector of the arc's flow rate, $\mathbf{A}$ and $\mathbf{\Lambda}$ are the $N \times J$ node-arc and the $K \times N$ tank-node incidence matrix, respectively.

The equations of the water storage and release in the water tank can be expressed by:

$$\mathbf{V}_{t+1} = \mathbf{V}_t + \mathbf{I}_t - \mathbf{O}_t \quad \forall t \quad (9)$$

$$0 \leq \mathbf{I}_t \leq \overline{\mathbf{I}} \quad \forall t \quad (10)$$

$$0 \leq \mathbf{O}_t \leq \overline{\mathbf{O}} \quad \forall t \quad (11)$$

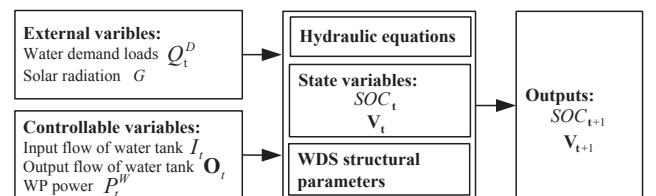


Fig. 2. Input-output relationship of the hydrodynamic model.

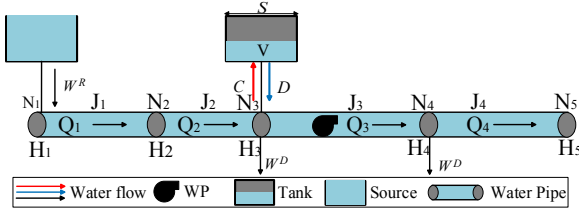


Fig. 3. Typical hydrodynamic network model of WECSS.

$$\underline{V} \leq V_t \leq \bar{V} \quad \forall t \quad (12)$$

$$SOC_t = (V_t - \underline{V}) / (\bar{V} - \underline{V}) \quad \forall t \quad (13)$$

$$V_t = \Gamma \Delta H_t \quad \forall t \quad (14)$$

where the storage water is influenced by the input/output flow of the tanks in (9) that is constrained by the maximum and minimum limits of the input flow, output flow and the storage water volume in (10)–(12) where $\bar{I}$ and $\bar{O}$ are the $K \times 1$ vectors of the maximum input and output flows, $\bar{V}$, $\underline{V}$ and V_t are the $K \times 1$ vectors of the maximum, minimum and real-time water storage in the tank. In (13), SOC_t represents the energy storage state. The pressure head at the tank nodes in (14) is determined by the water stored in the tanks where $H_t = (H_{t,1}, \dots, H_{t,N})$ is the pressure head and Γ is the $S \times S$ diagonal matrix of the tank surface parameters.

Based on the empirical Hazen-Williams head loss constraint in [39], the pressure head loss H_t due to the friction between the water flow and the pipe can be expressed by:

$$-A^T H_t = G \prod (Q_t) |Q_t|^{1.852} \quad \forall t \quad (15)$$

$$\underline{H} \leq H_t \leq \bar{H} \quad \forall t \quad (16)$$

$$-\bar{Q} \leq Q_t \leq \bar{Q} \quad \forall t \quad (17)$$

where G is the $F \times F$ diagonal matrix of the pipe friction parameters, $\bar{H}$ and $\underline{H}$ are the maximum and minimum pressure level, respectively, and Q_t is constrained in (17) by the minimum and maximum flow level $-\bar{Q}$ and $\bar{Q}$.

As an active component, WP can increase the pressure head and control the water flow rate which is expressed in (18)–(19).

Meanwhile, WP can only drive the water flow forward which is formulated in (20).

$$P_t^W = W_t^3 \odot (d1^U - e((1^U \odot W_t) \odot BQ_t)) \quad \forall t \quad (18)$$

$$BA^T H_t = W_t^2 \odot (a1^U - b((1^U \odot W_t) \odot BQ_t)^c) \quad \forall t \quad (19)$$

$$BQ_t \geq 0 \quad \forall t \quad (20)$$

where a , b , c , d and e are the WP parameters [40], $W_t = (W_{t,1}, \dots, W_{t,U})^T$ is the $U \times 1$ vector of the WP speed, B is the $U \times J$ WP-arc incidence matrix and 1^U is the $U \times 1$ unity vector.

The nodal pressure heads of the water sources such as reservoirs are considered to be constant, and are given by:

$$\Theta H_t = Y \quad \forall t \quad (21)$$

where Y is the $R \times 1$ vector representing the source nodes' elevation.

To summarize, (4)–(21) represent the detailed operation model of WECSS on multiple time-scales that is dispatched by the two-stage electric and hydraulic combined dispatching method introduced in Section 2.3.

2.3. Two-stage electric and hydraulic combined dispatching method

As observed from Fig. 4, a two-stage dispatching model including the day-ahead stage and the real-time stage was proposed to optimize WECSS operation in the smart community. In the day-ahead dispatching stage, the sub-problem of hydraulic dispatching to meet the

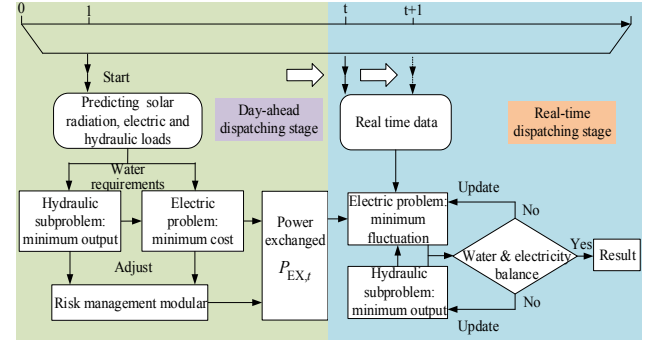


Fig. 4. Two-stage electric-hydraulic combined dispatching model.

predicted water load demand is addressed first and is followed by the optimization of WPs in the electric problem in order to minimize the cost. Originated from day-ahead dispatching, the power exchange between the community and the EG is applied to guide real-time dispatching. To eliminate the risk of SOC violation of WECSS due to forecast errors, a risk management modular is introduced to regulate the exchange power. In the real-time stage, based on the difference between the actual data and the predicted data, real-time MPC dispatching is devoted to mitigating the negative effects on both the electric sub-problem and the hydraulic problem. The corresponding detailed explanations are as follows:

Firstly, in the day-ahead dispatching stage, the water tank in the hydraulic side is optimized to release as little energy as possible with the given WP power:

$$F_{sp}: \min \sum_{t=1}^T O_t \quad (22)$$

$$\text{s.t. } SOC_T = SOC_0 \quad (23)$$

(8)–(21). where F_{sp} is the objective of the hydraulic sub-problem and O_t is the output flow.

Meanwhile, in the electric side, the objective function F_d is expected to minimize the electricity cost to the community by optimizing the WP power, as formulated by:

$$F_d: \min \sum_{t=1}^T (P_{EX,t}^{in} P_{in,t} - P_{EX,t}^{out} P_{out,t}) \quad (24)$$

$$\text{s.t. } \sum_{i=1}^{N_{PV}} P_{i,t}^{PV} + P_{EX,t} = \sum_{i=1}^{N_W} P_{i,t}^W + \sum_{i=1}^{N_E} P_{i,t}^{E-load} \quad \forall t \quad (25)$$

$$P_{EX,i} = P_{EX,j} \quad i, j \in \text{valley-hours} \quad (26)$$

where the power balance is expressed in (25), and for the EG, the energy consumption of the community in (26) is expected to be constant during the valley hours.

Secondly, as mentioned above, a risk management modular is introduced to ensure the reliability of the dispatching model. In other words, the SOC of WECSS should remain within [0 1] constraints for various risks, even when the actual data deviate $\epsilon\%$ (or $-\epsilon\%$) from the predicted value. As observed from Fig. 5, the exchange power can be

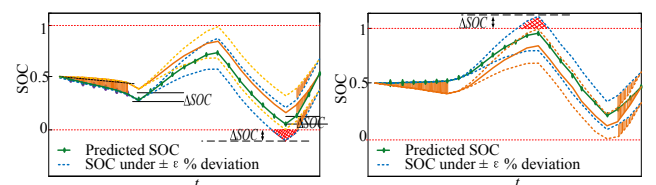


Fig. 5. Risk of SOC deviation from the upper or lower limits.

adjusted among the valley hours, as given by:

$$\begin{cases} \sum_{t=0}^m \Delta P_{EX,t}^{in} \geq \Delta SOC \cdot E_{WECSS}^{max} & m \in \text{valley} - \text{morning} \\ \sum_{t=22}^n \Delta P_{EX,t}^{in} \leq -\Delta SOC \cdot E_{WECSS}^{max} & n \in \text{valley} - \text{night} \end{cases} \quad (27)$$

where ΔSOC is the maximum deviation of SOC from the upper or lower limits, and E_{WECSS}^{max} is the maximum energy storage capacity of WECSS.

Although the electrical load of the smart community will be predicted and updated to the EG in the day-ahead stage, the exchange power between the community and the EG will fluctuate due to forecast errors.

Lastly, an MPC on the hydraulic and electric sides is conducted to mitigate the impact of forecast errors in the real-time stage. Assume that M is the prediction horizon of MPC and that $\wedge$ represents the predicted values. The hydraulic sub-problem in the real-time stage F_{rsp} is to minimize the decrease in the stored water.

$$F_{rsp}: \min \sum_{\tau=t}^{t+M-1} (-I_{\tau} + O_{\tau}) \quad (28)$$

$$\text{s.t. } SOC_{t+M-1} = \wedge SOC_{t+M-1} \quad (29)$$

(8)–(21).

Meanwhile, the electric problem in the real-time stage F_{rd} is focused on minimizing the power fluctuation.

$$F_{rd}: \min \sum_{\tau=t}^{t+M-1} |(P_{EX,\tau}^{in} - P_{EX,\tau}^{out}) - (\wedge P_{EX,\tau}^{in} - \wedge P_{EX,\tau}^{out})| \quad (30)$$

s.t. (5)–(7), (25).

2.4. Case study and parameters

A grid-connected MG with BIPV is shown in Fig. 1 and a typical WDS with 15 nodes, 14 arcs, 2 tanks, 1 water source and 3 water loads in Fig. 6 was introduced to verify the performance of WECSS. The detailed data of the WDS are provided in [41]. The typical curves of PV output, electric loads and water loads in the summer are shown in Fig. 7 [42–44]. The peak-valley electricity prices in China, i.e. 0.7 Yuan/kWh at 6:00–22:00 and 0.3 Yuan/kWh at 0:00–6:00 and 22:00–24:00 were adopted. The surplus electricity of MG is sold to the EG at 0.2 Yuan/kWh. The initial SOC of WECSS was set to 0.5.

Assuming that the dispatching strategy of WECSS proposed in the paper is Case 1 while WEISM (WISM & EISM) is Case 2 for comparison. In Case 2, the optimization objective function of WISM is used to minimize the operational cost as follows:

$$\min \sum_{t=1}^T \sum_{i=1}^{N_W} P_{i,t}^W \cdot P_{in,t} \quad (31)$$

s.t. (5)–(20).

The corresponding operational costs of EISM are given as follows:

$$\sum_{t=1}^T \left(\sum_{i=1}^{N_E} P_{i,t}^{E-load} P_{in,t} - \sum_{i=1}^{N_{PV}} P_{i,t}^{PV} \right) \quad (32)$$

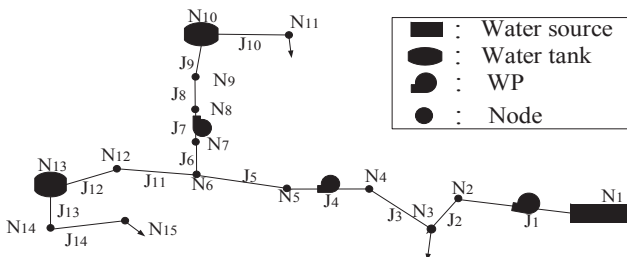


Fig. 6. Schematic representation of WDS.

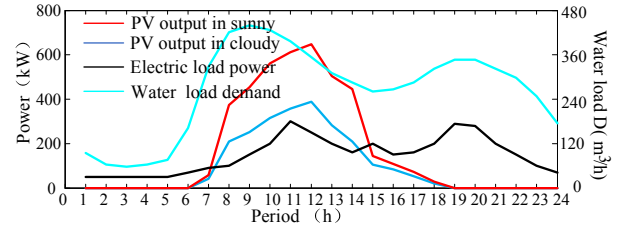


Fig. 7. Curves of PV output, electric load and water load.

2.5. Multiple integer linear programming formulation

Due to the nonlinear terms (such as (15), (18), (19) and (30)) and the complexity of the water-electricity combined dispatch, the optimization model is difficult to solve efficiently. An iterative procedure and linearized techniques were adopted to coordinate the novel dispatching system and meet the actual demand.

First of all, the optimization model with nonlinear terms is converted to a multiple integer programming problem.

For the nonlinear constraints, assume that $L_{x,t}(Q_{x,t}) = G_X \prod (Q_{x,t}) |Q_{x,t}|^{1.852}$ is the pressure head loss between two nodes with respect to (15). As shown in Fig. 8 (a), axis $Q_{x,t}$ is divided into $K-1$ segments, and the breakpoints are expressed by $q_x^1, q_x^2, \dots, q_x^K$. The breakpoint k is related to the continuous variable $\beta_{x,t}^k \in [0, 1]$ and the interval $[q_x^k, q_x^{k+1}]$ is related to the binary variable $u_{x,t}^k$.

$$Q_{x,t} = \sum_{k=1}^K q_x^k \beta_{x,t}^k \quad \forall x, \forall t \quad (33)$$

$$\hat{L}_{x,t} = \sum_{k=1}^K L_x(q_x^k) \beta_{x,t}^k \quad \forall x, \forall t \quad (34)$$

$$\sum_{k=1}^{K-1} u_{x,t}^k = 1 \quad \forall x, \forall t \quad (35)$$

$$\sum_{k=1}^K \beta_{x,t}^k = 1 \quad \forall x, \forall t \quad (36)$$

$$\beta_{x,t}^k \leq u_{x,t}^{k-1} + u_{x,t}^k \quad \forall x, \forall t, k = 2, \dots, K-1 \quad (37)$$

$$\beta_{x,t}^1 \leq u_{x,t}^1 \quad \forall x, \forall t \quad (38)$$

$$\beta_{x,t}^K \leq u_{x,t}^{K-1} \quad \forall x, \forall t \quad (39)$$

$$0 \leq \beta_{x,t}^K \leq 1 \quad \forall x, \forall t \quad (40)$$

where any given points $q_x^k \leq Q_{x,t} \leq q_x^{k+1}$ and $\hat{L}_{x,t}$ can be expressed by a linear combination of coordinates q_x^k and $L_x(q_x^k)$ with different weights $\beta_{x,t}^k$ [25] in (33) and (34). (35) ensures the only active line segment with the binary variable $u_{x,t}^k$ equal to 1. (36)–(40) represent the only pair of consecutive positive continuous variables, and $\beta_{x,t}^k$ and $\beta_{x,t}^{k+1}$ take non-zero values.

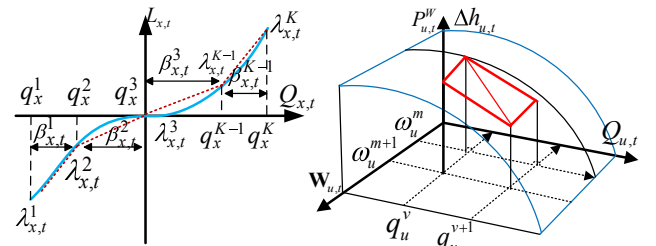


Fig. 8. Piecewise linear approximation of nonlinear functions.

Similarly, as shown in Fig. 8 (b), any given WP power $P_{u,t}^W(Q_{u,t}, W_{u,t})$ in (18) and pressure head $\Delta h_{u,t}(Q_{u,t}, W_{u,t})$ in (19) can be determined by the corresponding points $\hat{P}_{u,t}^W(q_u^m, \omega_u^m)$ and $\Delta \hat{h}_{u,t}(q_u^m, \omega_u^m)$ with different weights $\beta_{x,t}^k$ [25]. Limited to the space, no detailed description is given here.

For the nonlinear objective function, to minimize the fluctuation of exchange power in (30), the PV output error $\Delta P_{e-err,t} = \sum_{i=1}^{N_{PV}} P_{i,t}^{PV} - \sum_{i=1}^{N_{PV}} \hat{P}_{i,t}^{PV}$ should be mitigated by adjusting the WP's power as much as possible. Based on the analysis, taking a single period as an example, (30) can be given as follows if $\Delta P_{e-err,t} \geq 0$:

$$\min(-(P_{EX,t}^{in} - P_{EX,t}^{out}) + (\hat{P}_{EX,t}^{in} - \hat{P}_{EX,t}^{out})) \quad (41)$$

$$\text{s.t.} \quad \sum_{i=1}^{N_W} P_{i,t}^W \leq \sum_{i=1}^{N_W} \hat{P}_{i,t}^W + \Delta P_{e-err,t} \quad (42)$$

Similarly, for $\Delta P_{e-err,t} < 0$, the objective function and constraint are expressed by:

$$\min((P_{EX,t}^{in} - P_{EX,t}^{out}) - (\hat{P}_{EX,t}^{in} - \hat{P}_{EX,t}^{out})) \quad (43)$$

$$\text{s.t.} \quad \sum_{i=1}^{N_W} P_{i,t}^W \geq \sum_{i=1}^{N_W} \hat{P}_{i,t}^W + \Delta P_{e-err,t} \quad (44)$$

In addition, to speed up the solution procedure, an iterative framework was proposed for the first time to solve the hydraulic sub-problem and the electric problem. Taking the day-ahead stage as an example, the iterative procedure is described as follows.

Step 1: Initialization

The input parameters include loads, PV output and WDS data, and the evaluated WP power $P_{i,t}^{W*}$. Additionally, the initial iteration number $iter$ is set to 1, and the maximum iteration number is $Iter_max$, and the error is ε .

Step 2: Solve the hydraulic sub-problem

The objective (22) with constraints (8)–(21) and (23) can be transformed as follows:

$$F_{sp}: \min \gamma \quad (45)$$

$$\text{s.t.} \quad \gamma \geq \sum_t S_t \quad (46)$$

$$S_t \geq 0 \quad \forall t \quad (47)$$

$$\sum_{i=1}^{N_W} P_{i,t}^W - S_t = \sum_{i=1}^{N_W} P_{i,t}^{W*} \quad \forall t \quad (48)$$

(8)–(14), (16)–(21), (23), (33) and (40).

By solving the hydraulic sub-problem, the optimized objective γ and WP power $P_{i,t}^W$ can be obtained.

Step 3: Check stopping criteria

The solution procedure has two stopping criteria:

Rule 1: $iter > Iter_max$

Rule 2: $\gamma \leq \varepsilon$

If any of one of the criteria are satisfied, the iteration process will be terminated. If neither of the criteria are satisfied, update the $P_{i,t}^{W*} = P_{i,t}^W$ and go to Step 4.

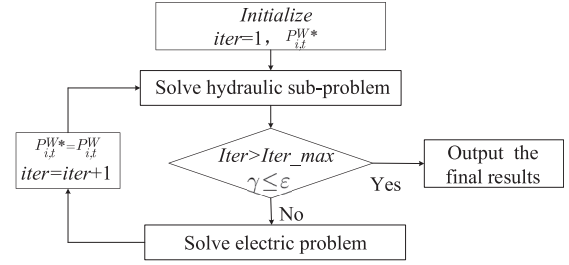


Fig. 9. Flowchart of the proposed iterative procedure.

Step 4: Solve the electric sub-problem

Solve the objective function (24) using the cut constraint $\sum_t \sum_{i=1}^{N_W} P_{i,t}^W = \sum_t \sum_{i=1}^{N_W} P_{i,t}^{W*}$ and Eqs. (5)–(7), and (25) and (26).

Step 5: Update $P_{i,t}^{W*} = P_{i,t}^W$ and return to Stage 2.

The solution process is illustrated in Fig. 9.

3. Results and discussion

3.1. Numerical simulations results

To facilitate the understanding of the performance of WECSS, the basic properties of WECSS were analyzed. On the one hand, the correlation between the flow rate, elevation difference and WP power is shown in Fig. 10 (a). As observed from the figure, with the pipe resistance parameter $G = 1297$, both the WP power and elevation difference between the nodes could drive the water flow increase while the rates of flow increase would decrease. For instance, the flow rate rose from 427 m³/h to 2998 m³/h as the WP power increased from 0 kW to 100 kW under the conditions of 25 m elevation difference, but it only increased to 3667 m³/h when the WP power reached 200 kW. As a consequence, WP determines the efficiency of the water supply. On the other hand, it is assumed that the WP is located in the front pipe, the tank is in the intermediate node, and the water load is in the later node, and the performance of the tank is shown in Fig. 10 (b). With 0 elevation difference and 1000 m³ water stored in the tank initially, it is observed that the input flow was controlled by the front WP, the output flow was determined by water demand and the flow difference was regulated by the tank. For instance, keeping 500 m³/h water demand fixed, the stored water increased to 2937 m³/h with a WP power of 100 kW while it was 3628 m³/h with a WP power of 200 kW. Consequently, the tank is essential for keeping the water supply and demand balance at all times.

Comparing Fig. 10 (a) and 10 (b), it can be concluded that the WP and tank are active components in WECSS. Fully considering their properties, the optimization results based on the proposed two-stage dispatching method are reported as follows:

First of all, the day-ahead electric consumption of WECSS and WEISM is shown in Fig. 11. As observed from Fig. 11(a), for WEISM on a sunny day, WDS consumed electricity regardless of the energy output

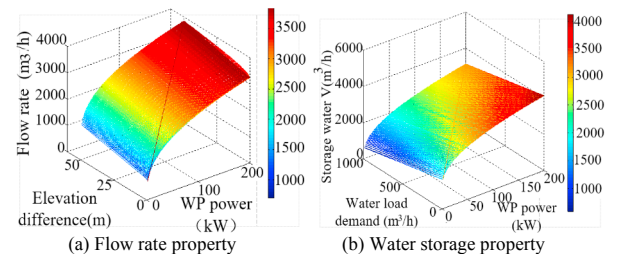


Fig. 10. Basic properties of WECSS.

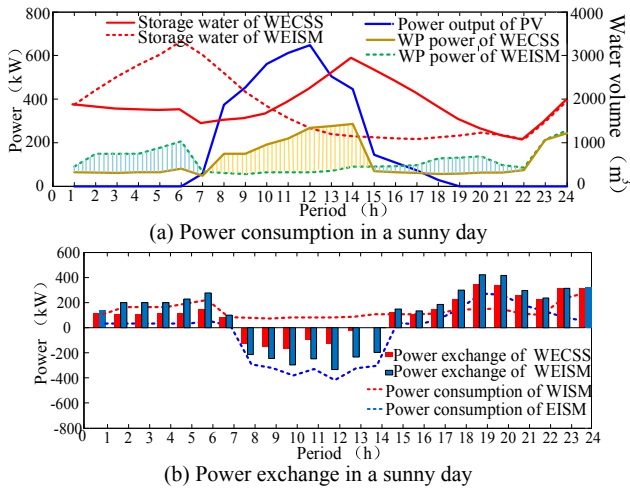


Fig. 11. Power consumption and power exchange in a sunny day.

of MG. Specifically, WPs did not follow the variation of PV output during the time period of 7:00–15:00 even if the abundant PV output exceeded the necessary power consumption of electric loads. In this case, the surplus power of MG was sold to the EG at a low price of 0.2 Yuan/kWh while WPs purchased power from the EG at a high price of 0.7 Yuan/kWh to maintain the necessary WDS operation, as shown in Fig. 11(b). This is because the price barriers block the direct deal between WISM and EISM when the EG must act as a middleman. To save operational costs, during the 0:00–6:00 and 23:00–24:00 time periods (valley hours), WDS consumed a high amount of electricity from the EG at a relatively low price of 0.3 Yuan/kWh to store as much water as possible. By comparison, for the WECSS with power and water supplied jointly, the power of WPs followed the change in the PV output when the PV power was sufficient with the increase in its SOC. During other periods, WPs maintained low-power operation while the water demand of the community was compensated by the water stored in the tanks in advance. Consequently, the target of peak shaving and valley filling was achieved.

As calculated by (24) and (31), (32), the expenditures of WECSS and WEISM were 1574 Yuan and 2899 Yuan (1398 Yuan for WISM, 1501 Yuan for EISM), implying that WECSS can save 1325 Yuan for property company and owners on a sunny day. Additionally, the maximum exchange power during the peak hours of the two cases was 421 kW and 347 kW, respectively, indicating that the community with WECSS has a lower and smoother power exchange and is more friendly to EG.

As shown in Fig. 12, compared with WEISM, the water supply load of WECSS during the peak hours was shifted to the valley hours in the cloudy day with insufficient PV output. The operational costs of WECSS and WEISM were 2115 Yuan and 3237 Yuan, respectively (1398 Yuan for WISM, 1839 Yuan for EISM), so that WECSS could save 1122 Yuan for the community. Additionally, the maximum exchange power of the two cases was 496 kW and 416 kW, respectively, implying that the MG with WECSS is even more friendly to EG.

Secondly, the robustness of WECSS under risks is shown in Fig. 13. As observed from Fig. 13(a), compared with the predicted value, the actual PV output would increase or decrease during the dispatching period on sunny days. Faced with the power deviation, the WP regulated their power consumption correspondingly, resulting in the changes in the storage energy in WECSS as shown in Fig. 13(b). It is observed from the figure that the SOC of WECSS under a 15% stochastic prediction error remained within the [0 1] constraint for various risks.

Consequently, with balancing the water supply and demand, WECSS

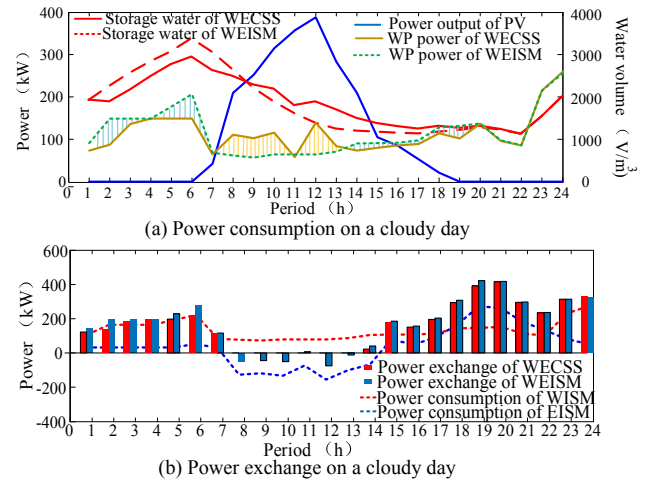


Fig. 12. Power consumption and power exchange on a cloudy day.

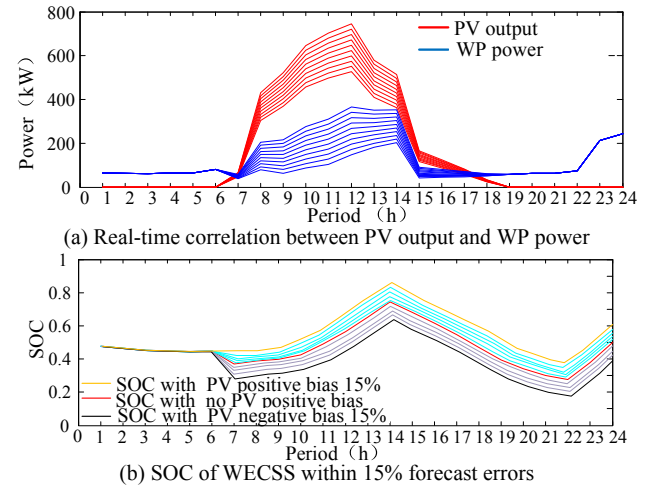


Fig. 13. Robustness analysis of WECSS.

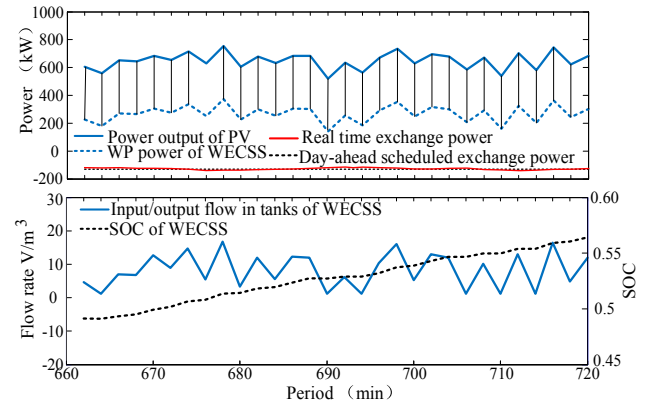


Fig. 14. Real-time electric and hydraulic balance recovery of WECSS.

has a perfect property to resist the influence of prediction errors. Note that for WEISM, due to the price barriers between WISM and EISM, the deviation of PV output caused by errors was transited to the EG while the water supply load maintained the same as predicted power.

Finally, the real-time electric-hydraulic balance is shown in Fig. 14.

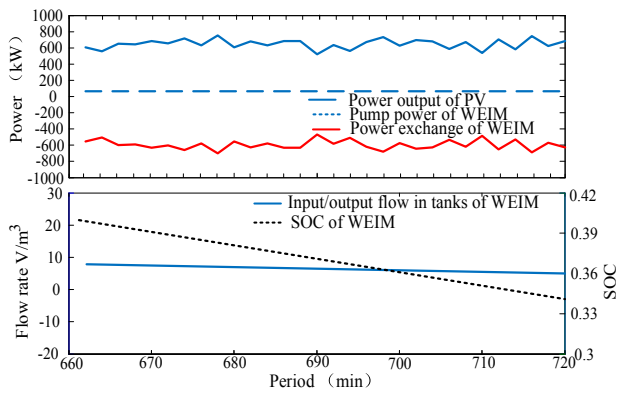


Fig. 15. Real-time electric and hydraulic balance of WEISM.

As observed from the figure, for WECSS, the WPs followed the fluctuation of PV output while water tanks regulated the input/output flow in the real-time stage. The electric balance and hydraulic balance were realized simultaneously. In addition, the red line in Fig. 14(a) suggests that real-time exchange power was almost the same as that dispatched in the day-ahead stage. However, for WEISM mode in Fig. 15, the fluctuation of the PV output was shifted to the exchange power because the WPs and the tanks did not respond. It can be seen that the WECSS makes it possible to mitigate the negative impact of solar radiation.

3.2. Economy analysis

For convenience's sake, assume the Spring and Autumn are relatively short while Winter and Summer are 180 days long, including 60% sunny days and 40% cloudy days. According to the data shown in Table 1, it can be calculated that WECSS with annual operational cost of 615,000 Yuan has a better economy than WEIM with 874,000 Yuan cost (WISM with 403,000 Yuan, EISM with 471,000 Yuan), which means that the annual operational cost for WECSS can be reduced by 29.6% (259,000 Yuan) compared to that of WEIM.

Furthermore, taking EISM as an example, as calculated by the Shapley value method (1)–(3), the cost distribution process is shown as Table 2. The annual operational cost of EISM was $235,500 + 106,000 = 341,500$ Yuan and the annual operational cost of WISM was $615,000 - 341,500 = 273,500$ Yuan, suggesting that the EISM and WISM can save 129,000 and 129,500 Yuan, respectively.

3.3. Comparison

To reflect the superiority of the two-stage dispatching model, a one-stage MPC dispatching model used in [35] was introduced to optimize the operational cost of WECSS. The dispatching results are shown as

Table 1
Operation cost for community users in summer.

Weather	Case	Electricity purchased (kWh)		Electricity sold (kWh)	Cost (Yuan)
		On-peak	Off-peak		
Sunny	WECSS	1866	1362	702	1574
	WEISM	1400	1391	0	1398
	EISM	3080	490	4010	1501
Cloudy	WECSS	2290	1705	0	2115
	WEISM	1400	1391	0	1398
	EISM	3080	490	2321	1839

Table 2

WECSS profit distribution based on the Shapley value method.

S	{EISM}	{WECSS}
$V(S)$	471,000	615,000
$V(S \setminus \{EISM\})$	0	403,000
$V(S) - V(S \setminus \{EISM\})$	471,000	212,000
$ S $	1	2
$W(S)$	1/2	1/2
$W(S)[V(S) - V(S \setminus \{EISM\})]$	235,500	106,000

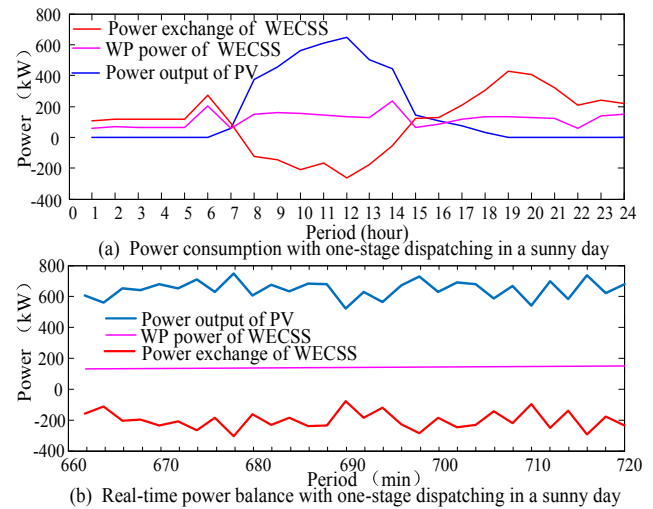


Fig. 16. Typical day-ahead and real-time curves with one-stage dispatching.

Table 3

Performance comparison between the iterative solution and direct solution.

Solution algorithms	Sunny		Cloudy	
	Cost (Yuan)	Time (sec)	Cost (Yuan)	Time (sec)
Iterative solution to the linearized model	1574	24.3	2115	25.1
Direct solution to the original model	1548	463.5	2076	519.7

Fig. 16. Comparing Fig. 16 (a) with Fig. 11 (a), it can be seen that the proposed dispatching model achieved peak load shaving and valley filling better than the one-stage model. As a result, the former with an operational cost of 1709 Yuan was more economical than the latter with 1574 Yuan. Further, comparing Fig. 16 (b) with Fig. 14 (a), the proposed two-stage model smoothed the exchange power and mitigated the negative impact of forecast errors in the real-time stage while the one-stage model failed.

In addition, the original model with nonlinear terms optimized directly by the COUENNE solver was presented for comparison with the proposed iterative solution to the linearized model by Gurobi. The operational costs and computation times for sunny and cloudy scenarios are shown in Table 3. Compared with the iterative solution to the linearized model, the direct solution to the original model obtained lower operating costs for WECSS in both scenarios, accounting for 1.4% and 1.2% respectively. However, the computation times of the original model was more than 19 times longer. The computational results show that the proposed solution algorithm provides fairly accurate results in a relatively short time.

4. Conclusion

In this paper, a novel water-electricity combined supply system with economy and universality involving different stakeholders was proposed to solve the problems arising from the integration of building-integrated photovoltaics. To evaluate the performance of the water-electricity combined supply system on multiple time-scales, a detailed electric-hydraulic combined analytic model incorporating the power flow limits and WP ramp rate and a two-stage dispatching method were proposed, followed by the iterative solution algorithm to efficiently solve the optimization model. The simulation study indicated that the water-electricity combined supply system could ensure the balance between supply and demand of both water and electricity, and based on the Shapley value method, the fair distribution of cooperative profits could be realized.

Based on the proposed two-stage dispatching method, the water-electricity combined supply system could achieve peak shaving and valley filling and smooth power fluctuation by storing the water in tanks and regulating the operating power of water pumps. Therefore, the operational cost of the community could be reduced by 29.6% (259,000 yuan) while the operational pressure of the external grid could be decreased. In addition, the proposed iterative algorithm could greatly improve the computational efficiency of the operation model for a water-electricity combined supply system by more than 19 times.

Although the excellent performance of the water-electricity combined supply system on multiple time-scales was verified by the simulation, the work was only about theoretical discussion and ignored some technical details. In real application, the water tanks should be equipped with the self-regulating valves to flexibly control the water inflow and outflow while the water pumps should have the function of automatic power regulation and both of them should be able to respond to remote control instructions. Future work also will include the experimental verification of the model results.

Acknowledgments

This work was supported in part by the Science and Technology Project of Guangxi Power Grid Co., Ltd., China Southern Power Grid (GXKJXM20170244), in part by the National Natural Science Foundation of China (51537003). Also, we gratefully acknowledge the technical assistance and the insightful idea from China Scholarship Council “International Clean Energy Talent Program (iCET)”.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2019.04.059>.

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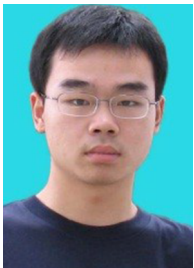
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